

Global Earthquake Explorer - Help Documentation

STA5092Z - Exploratory Data Analysis, Assignment 3

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Contents

1	Overview	2
2	Setup and Running the App	2
2.1	Prerequisites	2
2.2	File Structure	2
2.3	Launching the App	2
3	App Structure	2
4	Tab-by-Tab Guide	3
4.1	Feature Distributions	3
4.1.1	Sidebar Controls	3
4.1.2	Output Panels	3
4.1.3	Worked Example	3
4.2	Temporal Trends	3
4.2.1	Sidebar Controls	3
4.2.2	Output Panels	4
4.2.3	Worked Example	4
4.3	Spatial Explorer	4
4.3.1	Sidebar Controls	4
4.3.2	Output Panels	4
4.3.3	Worked Example	5
4.4	Top Earthquakes	5
4.4.1	Sidebar Controls	5
4.4.2	Output Panels	5
4.4.3	Worked Example	5
4.5	About	5
5	Data Notes	5
6	References	6

1 Overview

The Global Earthquake Explorer is an interactive R Shiny application built to explore global earthquake data spanning from January 1965 to March 2023. The app allows users to investigate the distributions, temporal trends, and spatial patterns of earthquakes across a range of magnitudes, depths, and time periods. The application was developed using R and the Shiny framework, with visualisations produced via ggplot2. The data are the same cleaned and merged datasets used in Assignment 2.

2 Setup and Running the App

2.1 Prerequisites

Ensure the following R packages are installed before running the app:

- shiny
- tidyverse
- lubridate
- maps
- ggplot2
- DT

Install all at once with:

```
install.packages(c("shiny", "tidyverse", "lubridate", "maps", "ggplot2", "DT"))
```

2.2 File Structure

The app expects the following directory structure:

```
MLNBUN001/  
  MLNBUN001.R  
  data/  
    Earthquakes 1965 - 2016.csv  
    query.csv
```

Important: Both CSV files must reside inside a subdirectory called `data/` in the same folder as `MLNBUN001.R`. The files should not be edited or renamed.

2.3 Launching the App

Open `MLNBUN001.R` in RStudio and click “Run App”, or run from the R console:

```
shiny::runApp("MLNBUN001.R")
```

3 App Structure

The app is organised into five tabs, accessible via the top navigation bar:

Tab	Purpose
Feature Distributions	Explore variable distributions and scale composition
Temporal Trends	Visualise earthquake frequency over time
Spatial Explorer	Interactive map with region zoom options
Top Earthquakes	Searchable table and map of the largest events
About	Data sources, processing notes, and references

4 Tab-by-Tab Guide

4.1 Feature Distributions

This tab explores the distribution of magnitude and depth across the filtered dataset.

4.1.1 Sidebar Controls

- **Year Range** - Slider to restrict the analysis to a specific time window (1965-2023).
- **Magnitude Range** - Slider to select a range of earthquake magnitudes to include.
- **Scale Categories** - Checkboxes to select which Richter-scale categories to display (e.g. Moderate, Strong, Major, Great, Extreme).
- **Feature to plot** - Dropdown to switch between “Magnitude” and “Depth (km)” distributions. Depth is plotted on a log scale to handle its skewed distribution.
- **Number of bins** - Adjusts the histogram resolution.

4.1.2 Output Panels

- **Histogram (left)**: Shows the count of earthquakes across the selected feature, coloured by Richter scale category using a consistent cool-to-warm colour palette.
- **Scale Composition Pie Chart (right)**: Shows the proportional breakdown of scale categories in the filtered data.
- **Summary Statistics Table (below)**: Reports the sample size, and minimum, mean, and maximum values for both magnitude and depth.

4.1.3 Worked Example

To see the depth distribution of major and great earthquakes since 2000:

1. Set Year Range to 2000-2023.
2. Set Magnitude Range to 7.0-10.0.
3. Check only Major and Great under Scale Categories.
4. Select Depth (km) in the “Feature to plot” dropdown.
5. The histogram reveals that most major/great earthquakes occur at relatively shallow depths.

4.2 Temporal Trends

This tab investigates how earthquake frequency has changed over time.

4.2.1 Sidebar Controls

- **Year Range** - Restricts both temporal plots to the selected period.

- **Scale Categories** - Filters the stacked bar chart to the selected categories.
- **LOESS Smoothing Span** - Controls the smoothness of the trend line in the M5+ plot. Lower values produce a more flexible, wiggly fit; higher values produce a smoother trend.

4.2.2 Output Panels

- **Annual Earthquake Counts by Scale (top)**: Stacked bar chart showing the number of earthquakes per year, broken down by Richter scale category. The noticeable rise in lower magnitude events from the late 1990s reflects improved global seismic monitoring rather than a true increase in activity.
- **Magnitude 5.0+ Trend (bottom)**: A line plot of annual M5+ earthquake counts overlaid with a LOESS smoothing curve (red) and a 95% confidence band (orange shading). This plot shows the long-term trend is relatively stable, supporting the monitoring-bias interpretation.

4.2.3 Worked Example

To investigate whether strong and major earthquakes have become more frequent after 2000:

1. Set Year Range to 1990-2023.
2. Select only Strong and Major under Scale Categories.
3. The stacked bar chart shows these categories remain relatively stable across years, consistent with the monitoring-bias hypothesis.
4. Increase the LOESS span to 0.6 for a smoother M5+ trend line.

4.3 Spatial Explorer

This tab provides an interactive map of earthquake locations with flexible filtering and region-zoom options.

4.3.1 Sidebar Controls

- **Minimum Magnitude** - Only earthquakes at or above this magnitude are plotted.
- **Year Range** - Restricts plotted events to the selected time window.
- **Select Region** - Zoom into one of four pre-defined regions, or define a custom bounding box:
 - Global - Full world map.
 - Western Pacific (Ring of Fire) - Lon: 100 to 180, Lat: -60 to 70.
 - Mediterranean Belt - Lon: -15 to 65, Lat: 25 to 50.
 - South American Subduction Zone - Lon: -85 to -30, Lat: -60 to 15.
 - Custom - Enter any bounding box using the four numeric inputs that appear.
- **Colour by** - Map either Magnitude or Depth to the point colour gradient (orange to dark red).
- **Point Transparency** - Adjusts alpha to manage overplotting in dense regions.
- **Max Point Size** - Scales the maximum size of plotted points.

4.3.2 Output Panels

- **Map (main)**: ggplot2 map with earthquake locations plotted as points. Colour and size both encode the selected variable/magnitude, making high-severity events visually prominent.
- **Summary Statistics (below map)**: Counts and magnitude statistics for all events visible in the current filtered view.

4.3.3 Worked Example

To explore high-magnitude events in the Mediterranean since 2000:

1. Set Minimum Magnitude to 5.0.
2. Set Year Range to 2000-2023.
3. Select Mediterranean Belt from the Region dropdown.
4. Set Colour by to Depth (km) to see whether shallow or deep events dominate.
5. The map reveals concentrated activity in Turkey, Greece, and Iran.

4.4 Top Earthquakes

This tab displays a searchable, sortable table of the largest earthquakes in the filtered dataset, alongside a world map showing their locations.

4.4.1 Sidebar Controls

- **Number of top earthquakes** - Slider to choose how many of the largest events to display (5 to 50).
- **Year Range** - Filter events to a specific period.
- **Minimum Magnitude** - Only events at or above this value are eligible for inclusion.

4.4.2 Output Panels

- **Table (top)**: An interactive DataTable showing date, latitude, longitude, depth, magnitude, and scale category for the selected top earthquakes. The table supports searching, sorting, and pagination.
- **Map (bottom)**: A world map with the top earthquakes plotted and labelled with their magnitude (e.g. “M9.1”). Point colour and size both encode magnitude.

4.4.3 Worked Example

To find the 10 largest earthquakes since 2004:

1. Set Number of top earthquakes to 10.
2. Set Year Range to 2004-2023.
3. Set Minimum Magnitude to 7.0.
4. The table shows the 2004 Sumatra (M9.1) and 2011 Tohoku (M9.1) events at the top.
5. The map clearly shows these are concentrated along Pacific subduction zones.

4.5 About

The About tab provides a brief description of the app, the data sources, processing steps, and references. No interactive controls are available on this tab.

5 Data Notes

- The combined dataset contains 34,168 earthquake observations after filtering for type = “Earthquake”.
- Core variables (magnitude, depth, latitude, longitude, datetime) contain no missing values.

- Uncertainty variables (`depth_error`, `magnitude_error`, `horizontal_error`, `azimuthal_gap`) contain substantial missing data and are not used in visualisations.
- Magnitude scale types differ between the two source datasets (`mb/ml` vs `mww`); cross-period magnitude comparisons should be interpreted with caution.

6 References

1. Gutenberg, B., & Richter, C. F. (1956). Bulletin of the Seismological Society of America, 46(2), 105-145.
2. Isacks, B., Oliver, J., & Sykes, L. R. (1968). Journal of Geophysical Research, 73(18), 5855-5899.
3. Jackson, J. A., & McKenzie, D. (1984). Geophysical Journal of the Royal Astronomical Society, 77(1), 185-264.
4. Lay, T., & Wallace, T. C. (1995). Modern Global Seismology. Academic Press.